VINOTH RAJ R

AI Engineer · Software Engineer, AI Automation, CRM & Mobile Platforms

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SUMMARY

Software engineer with 3.5+ years shipping production systems, now building the AI layer of a financial-services sales operation end to end. At AssetPlus I set up and run the full CRM, replaced tele-calling with an AI voice agent and a root-free Android call-automation app, put LLMs to work reviewing every call, and built the marketing and business-metrics dashboard leadership uses daily. I also publish open-source AI tooling (**ai-knowhow**, **llm-json**). Stack: Python, Frappe/ERPNext, Flutter + Kotlin, React/Next.js, and LLMs (Sarvam AI, Gemini, OpenAI, Claude).

EXPERIENCE

Software Engineer · AssetPlus

Apr 2025 – Present

Chennai, Tamil Nadu · On-site · Financial services (mutual-fund distribution)

Full CRM setup and ownership (Frappe CRM)

- Set up and own the company's **Frappe CRM end to end**: lead capture from Meta Instant Forms and the website, AI-driven lead prioritisation, Zoho sync, Google Calendar meeting scheduling, and WhatsApp/Slack alerts, with Frappe Insights dashboards for team performance and lead conversion.
- Integrated the **Meta Conversions API**: when a lead registers, the CRM sends the conversion back to Meta so ad delivery optimises for real registrations, not form fills.
- Built WhatsApp automation (WPPConnect + Puppeteer) inside Frappe for instant hot-lead alerts and automated group onboarding; tuned real-time lead sync and message queues.

AI calling and AI call QA

- Designed an **AI voice-agent pipeline on Sarvam AI** that calls every new ad lead and does first-level screening before a human dials: leads created 10 AM–7 PM are called immediately, night leads batch to 10 AM, and failed calls retry after 2 days but never on Sunday. Outcomes drive automatic prioritisation and push hot leads to the sales team's WhatsApp groups.
- Built a **server-side LLM QA pipeline**: Sarvam transcription, then four Gemini prompts (summary, tags, scoring metrics, QA questions) on every recorded call, stored against the lead. Added output normalisation that removed parse failures, token and cost tracking, and an LLM benchmark harness (Gemini vs OpenAI) to choose models on evidence.

Call automation mobile app (Flutter + Kotlin)

- Replaced traditional tele-calling with an in-house Android app that captures every SIM and WhatsApp call log and recording into the CRM. Re-engineered it from a rooted custom-ROM build to a **root-free** app (Shizuku + scrcpy audio capture) that runs on the phones agents already carry.
- Raised recording-upload reliability from **92% to 100%** with zero duplicates: offline-first queue, idempotent server upserts, hourly backfill, and a fix for a concurrent-checkpoint race.
- Cut app-open time at 1,000 calls from **19.1 s to 3.9 s** by moving recording lookup off the UI thread into one batched native index. Recordings are now written as mono 16 kHz in real time, about 6× smaller on disk.
- Added device-health monitoring: event-driven status reports (recording service up/down, missing audio, 2-hour heartbeat in working hours) so the server can alert ops on WhatsApp the moment a device stops recording. Per-status de-duplication cut redundant status records from 13 to 0 per restart cycle.
- Shipped agent-facing features: utilisation logging with server approval sync, talk/ring-time dashboards, Excel export, and an in-app diagnostics log.

Marketing and business metrics

- Built the **Har Ghar MFD analytics dashboard** (React + Vite on Vercel) used by management: GA4 (via Windsor.ai), Vimeo, Meta and Google Ads spend, Microsoft Clarity, and CRM leads-to-deals in one place, refreshed daily by GitHub Actions with a data-verification step.
- Reports cost per lead by campaign, platform-claimed vs CRM-received leads, retention, content and video performance, and benchmark cards against cited industry figures. An admin-gated **AI Insights tab (Gemini)** writes executive summaries and answers questions from a bounded data brief.
- Found that 166 of 167 Google Ads leads reached the CRM without a campaign (₹1.69 lakh of spend unattributable) and specified the UTM/gclid capture fix and a Frappe lead-and-deal API to close the loop.

AI enablement

- Built a public AI knowledge assistant on Dify (Next.js frontend, server-side key isolation, rate limiting, SSE streaming, two embed modes) designed for 1,000+ users a day on free-tier hosting.
- Rolling out my open-source **ai-knowhow** skills company-wide so every department picks the right AI model and effort for the job and learns how AI-built work was done.

Senior Full Stack Engineer · Makto Technology Dec 2022 – Mar 2025

Coimbatore, Tamil Nadu · Remote · UAE operations

- Built a mobile-first delivery platform in Flutter: customer and delivery-agent apps with Firebase real-time data, Stripe payments and Google Maps live tracking, plus a React admin panel and warehouse management system (role-based access, inventory, master data over REST).
- Led **5+ ERPNext implementations** across CRM, fleet, healthcare and education; shipped 3 internal ERPNext apps that cut operational costs by **40%** and lifted team productivity by **50%**.
- Automated data collection by integrating LinkedIn and Google Forms with ERPNext; set up Git-based CI/CD; active contributor to the Frappe developer community.

OPEN SOURCE

ai-knowhow MIT · Agent Skills for Claude Code, claude.ai, Codex and ChatGPT · github.com/VINOTH-RAJ-R/ai-knowhow

- **modelwise** turns a basic prompt from any department into a top-practitioner plan, the right model and reasoning effort per phase, a cost comparison across models, and a prompt written for that model, driven by a validated registry of 61 Anthropic and OpenAI models. **modelwise-run** then executes the task on the chosen Claude model after confirmation.
- **retrace** turns a finished AI-assisted task into a saved lesson (decisions, prompts that worked, mistakes, the cheaper path) and teaches it. Plain Markdown, JSON and one dependency-free Python validator; no server, no data collection.

llm-json MIT · Python 3.9+ · zero runtime dependencies · CI · github.com/VINOTH-RAJ-R/llm-json

- Extracts a usable object from LLM output (code fences, preambles, truncated or single-quoted JSON) through staged, opt-in repairs, and on failure raises with the raw response, the stage it stopped at and caller context attached, so production parse errors are debuggable.

TECHNICAL SKILLS

AI & LLMs	Sarvam AI (voice agents, STT), Gemini, OpenAI, Claude, Dify / RAG, prompt pipelines, agent skills, LLM benchmarking, structured-output parsing, token & cost tracking
Backend & CRM	Python, Frappe / ERPNext, Frappe CRM, REST APIs, webhooks, cron / Redis queues, MySQL, PostgreSQL, Zoho, Meta Conversions API
Mobile	Flutter (Dart), Kotlin, Android foreground services, telephony, Shizuku, Provider, Dio, offline-first sync
Frontend & Analytics	React, Next.js, TypeScript, Vite, GA4, Windsor.ai, Meta & Google Ads data, Microsoft Clarity, Vimeo API, Frappe Insights
Automation	WhatsApp (WPPConnect, Puppeteer), Slack, Google Calendar API, Stripe, Google Maps, Firebase
DevOps & Tools	Git, GitHub Actions, CI/CD, Vercel, GitHub Pages, Linux, Claude Code

EDUCATION & PROFILES

Bachelor of Information Technology, Sona College of Technology, Salem · Jun 2022 · CGPA 7.64

LeetCode · Stack Overflow · Frappe Community